

Clinical Decision Support Systems (CDSS) for OCT Data: What They Are and How to Build One

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A clinical decision support system turns patient data into timely, actionable guidance for a clinician at the moment of a decision. This document explains what a CDSS is, how one is architected, what an OCT-based CDSS looks like (using dental OCT as the running example), and the full path to developing, validating, and deploying one.

1. What a CDSS is

A **clinical decision support system (CDSS)** is software that takes information about a specific patient, combines it with clinical knowledge, and presents patient-specific assessments or recommendations to a clinician to support a decision. The key word is *support*: the clinician remains responsible for the decision.

A CDSS is different from a standalone AI model. A model outputs a prediction (“87% probability of dentin caries”). A CDSS wraps that prediction in everything needed to make it clinically usable: data intake, quality checks, clinical rules and guidelines, explanation, uncertainty, a recommendation phrased in clinical terms, a user interface, and integration into the clinician’s workflow and records.

Two broad families

- **Knowledge-based CDSS** — encodes expert knowledge as explicit rules (“IF lesion crosses the dentin–enamel junction AND lesion is active THEN consider restoration”). Transparent and easy to audit, but brittle and cannot learn patterns from images.
- **Non-knowledge-based (data-driven / ML) CDSS** — learns patterns from data, typically with deep learning on images. Powerful on imaging, but opaque, data-hungry, and prone to failure on unfamiliar data.
- **Hybrid** — the practical modern design, and the right one for OCT: a deep-learning model extracts measurements from the image (lesion depth, gap size, layer thickness), and a transparent rule layer maps those measurements onto guideline-based recommendations.

Passive vs active

- **Passive** — the clinician opens the tool and asks for its output.
- **Active** — the system pushes alerts or suggestions automatically (e.g. flagging a suspicious region as soon as a scan is taken). Active systems carry a risk of **alert fatigue** if they fire too often.

Levels of autonomy

- **Assistive (second reader)** — highlights findings; the clinician makes every call. The most common and easiest to validate and regulate.
- **Augmentative / triage** — prioritises or quantifies (e.g. measures lesion depth, sorts urgent cases first).
- **Autonomous** — makes a decision without clinician review (rare; exists for diabetic retinopathy screening from fundus photos). Requires the strongest evidence.

For a new modality like dental OCT, an **assistive or augmentative hybrid CDSS** is the realistic and defensible target.

2. How a CDSS works: the architecture

An imaging CDSS is a pipeline. Each layer has a distinct job, and most real-world failures happen in the layers around the model, not in the model itself.

1. **Data acquisition** — the scan is captured by the device (OCT volume or B-scans), plus relevant patient context (age, history, previous scans, clinician notes).
2. **Ingestion and standardisation** — images arrive in a known format with metadata (device, protocol, resolution, timestamp), are de-identified where needed, and are converted to a consistent spacing and intensity range.
3. **Quality control gate** — checks whether the scan is usable at all: signal strength, motion artifacts, saliva or blood obstruction, wrong anatomy, out-of-focus. Unusable scans are rejected with a reason (“rescan: low signal”) rather than analysed. This single layer prevents many dangerous outputs.
4. **Perception / inference engine** — the ML models: segmentation (enamel, DEJ, dentin, lesion, restoration gap), detection, and classification. Output is measurements and probabilities, not recommendations.
5. **Uncertainty and out-of-distribution (OOD) check** — estimates how confident the model is and whether the input resembles the training data. High uncertainty or OOD input triggers abstention: “unable to assess reliably, clinician review required.”

6. **Knowledge / reasoning layer** — turns measurements into clinical meaning using guidelines and rules, e.g. lesion confined to outer enamel → non-invasive management; lesion into dentin with cavitation → restorative consultation. Keeping this layer explicit makes the system auditable and easy to update when guidelines change.
7. **Explanation layer** — shows *why*: overlays of the segmented lesion on the scan, the measured depth, the rule that fired, and similar reference cases. Explanations should be anatomically grounded (a measured lesion boundary) rather than only heatmaps.
8. **Presentation / user interface** — delivers the output in the clinician's language, at the right moment, with minimal clicks: finding, severity, confidence, recommendation, and the option to accept, edit, or reject.
9. **Integration** — connects to the practice management system or electronic health record so results are stored with the patient record and can be compared over time.
10. **Feedback and monitoring** — logs clinician overrides, tracks performance and data drift after deployment, and feeds curated cases back for model updates under change control.

A compact way to remember it: **input → quality → perception → uncertainty → reasoning → explanation → interface → integration → monitoring.**

3. What an OCT-based CDSS looks like

OCT is well suited to decision support because it gives **quantitative, depth-resolved structure**: instead of “there may be decay”, it can say “the lesion extends 0.9 mm deep and has crossed the dentin–enamel junction.” Measurements like that map naturally onto clinical thresholds.

Candidate decisions for a dental OCT CDSS

Clinical decision	What OCT measures	Possible recommendation
Monitor, remineralise, or restore a carious lesion	Lesion depth, integrated reflectivity, DEJ involvement, surface integrity	Fluoride/sealant and recall vs restorative treatment
Keep or replace an existing filling	Marginal gap length, debonding, secondary caries under margin	Monitor vs repair vs replace

Clinical decision	What OCT measures	Possible recommendation
Is a lesion progressing?	Change in depth or reflectivity between visits	Escalate or de-escalate management
Periodontal status	Gingival thickness, sulcus, subgingival calculus	Hygiene or periodontal referral
Oral mucosal lesion triage	Epithelial thickness, basement-membrane integrity	Routine review vs urgent biopsy referral
Crack assessment	Crack presence and depth	Monitor vs protective restoration

Pipeline for the flagship use case (caries management)

1. Clinician scans the tooth surface with an intraoral OCT probe.
2. Quality gate checks signal, motion, and that the target surface is in view.
3. Segmentation model delineates enamel surface, lesion, DEJ, and dentin in each B-scan, then aggregates across the 3D volume.
4. Measurement module computes maximum lesion depth, depth relative to enamel thickness, lesion volume, reflectivity, and surface cavitation.
5. Uncertainty module scores confidence; out-of-distribution scans are flagged.
6. Rule layer maps measurements onto a caries management framework (e.g. ICDAS-style staging linked to non-invasive vs operative treatment).
7. Interface shows the overlay, the numbers, the confidence, and the suggested management, and stores it for comparison at the next visit.

Example output (illustrative)

Tooth 36, occlusal surface. Subsurface lesion detected. Maximum depth 1.1 mm; extends ~0.2 mm beyond the dentin–enamel junction. Surface appears intact (non-cavitated). Model confidence: high. **Suggested:** discuss minimally invasive management (e.g. sealant or resin infiltration) with review in 6 months; compare with baseline scan. *Clinician to confirm.*

Why this beats a plain classifier

A classifier says “dentin caries: 0.87.” A CDSS explains where, how deep, how sure, what that means clinically, what to do next, and when not to trust itself — and it tracks the same tooth over time.

4. How to develop one: a phased roadmap

Phase 0 — Define the decision with clinicians

Start from a decision, not a model. With dentists, write down: the exact clinical question, who uses the output and when, what they currently do, what error is worse (missing a lesion vs over-treating), and what a useful output looks like. Record this as an **intended-use statement** — it drives data collection, metrics, and later regulation.

Phase 1 — Data and ground truth

- Secure a data source: a dental school or clinic partnership with ethics approval and a data-sharing agreement.
- Decide the protocol: device, wavelength, scan pattern, which teeth and surfaces, ex vivo (extracted teeth) first, then in vivo.
- Define ground truth: micro-CT or histology for extracted teeth; expert consensus (multiple dentists, adjudicated) for in vivo; histopathology for mucosal lesions.
- Write an **annotation guide** with clear definitions and examples; measure inter-annotator agreement.
- Split data **by patient/tooth**, never by B-scan, to avoid leakage. Reserve an external test set from a different device or site if at all possible.

Phase 2 — Model development

- Preprocessing: speckle handling, intensity normalisation, surface flattening/alignment.
- Perception models: segmentation (U-Net / nnU-Net family, transformer variants) and 3D volume models where data allows.
- Label efficiency: self-supervised pretraining (possibly on retinal OCT, which shares physics), augmentation, semi-supervised learning.
- Measurement module: derive clinically meaningful quantities from the segmentation.

Phase 3 — Safety layers

- Image-quality classifier for the quality gate.
- Uncertainty estimation (ensembles, Monte Carlo dropout, conformal prediction) and calibration.
- Out-of-distribution detection and a defined abstention policy.
- Explicit rule layer linking measurements to recommendations, reviewed by clinicians.

Phase 4 — Interface and workflow

- Prototype the screen with clinicians; iterate on what they actually read in the seconds they have.
- Design for accept / edit / reject, and log every override.
- Plan integration with practice software and longitudinal comparison.

Phase 5 — Validation ladder

1. Internal test performance.
2. External validation on another site or device.
3. **Reader study**: dentists with vs without the CDSS on the same cases — does it improve their decisions?
4. Prospective, real-world study in clinic.

Phase 6 — Deployment and monitoring

Version-controlled models, performance and drift monitoring, incident reporting, and a controlled process for updates.

A realistic research scope for a first paper covers Phases 0–3 plus an external test and a small reader study — that alone would be a substantive contribution in dental OCT.

5. Evaluation and validation

A CDSS is judged on whether it improves decisions safely, not on accuracy alone.

Model-level metrics

- **Sensitivity and specificity** at a clinically chosen operating point — report both, and justify the threshold by which error is costlier.

- **AUROC / AUPRC** — threshold-free discrimination; AUPRC matters when lesions are rare.
- **Segmentation metrics** — Dice, Hausdorff distance, and above all **measurement error** (e.g. mean absolute error of lesion depth in micrometres against micro-CT), because the measurement drives the decision.
- **Agreement** — Cohen's kappa or ICC between the system and experts, compared with agreement between experts themselves.

Reliability metrics

- **Calibration** — does “80% confident” mean right 80% of the time? Report expected calibration error and reliability diagrams.
- **Selective prediction** — risk-coverage curves: how error falls as the system abstains on its least certain cases, and how many cases it defers.
- **OOD detection** — AUROC for separating in-distribution scans from other devices, poor-quality scans, or unseen conditions.
- **Subgroup performance** — by tooth type, surface, age group, device, and site.

Decision-level metrics

- **Decision curve analysis / net benefit** — whether acting on the CDSS output does more good than harm across plausible thresholds, compared with “treat all” and “treat none.”
- **Treatment concordance** — how often the suggested management matches an expert panel.

Human-AI evaluation

- **Reader studies** — clinicians assess the same cases with and without the CDSS (ideally crossover, with a washout). Measure accuracy, confidence, time per case, and inter-reader variability.
- Watch for **automation bias** (following wrong advice) and **under-reliance** (ignoring right advice).

Reporting guidelines to follow

- **CLAIM** — checklist for AI in medical imaging.
- **TRIPOD+AI** — prediction model development and validation.

- **STARD-AI** — diagnostic accuracy studies.
- **DECIDE-AI** — early-stage live clinical evaluation of decision support.
- **CONSORT-AI / SPIRIT-AI** — randomised trials and their protocols.

Following these from the start makes the work far easier to publish and far more credible.

6. Regulation, standards, and data governance

A research prototype does not need clearance, but designing with regulation in mind from day one saves a rebuild later. Rules evolve, so confirm current requirements before any commercial step.

Is it a medical device?

Software that analyses patient images to support diagnosis or treatment decisions is generally **Software as a Medical Device (SaMD)**.

- **EU (relevant for Ireland)** — the **Medical Device Regulation (MDR)**; under its software classification rule, decision-support software is typically **Class IIa or higher**, requiring a notified body. The **EU AI Act** additionally treats AI that is a regulated medical device as **high-risk**, adding requirements on data governance, transparency, human oversight, and logging.
- **US** — FDA pathways such as **510(k)** (with a predicate device) or **De Novo** (novel, moderate risk). Some clinical decision support that lets clinicians independently review the basis of a recommendation falls outside device regulation, but image-analysis software generally does not.

Core standards

- **ISO 13485** — quality management system for medical device makers.
- **ISO 14971** — risk management: identify hazards (missed lesion, over-treatment, wrong tooth) and mitigations.
- **IEC 62304** — software lifecycle for medical device software.
- **IEC 62366** — usability engineering, including use errors.
- Good machine learning practice principles (published jointly by FDA, Health Canada, and the UK MHRA) — data representativeness, independent test sets, human-AI team performance, monitoring.

Data protection and ethics

- **GDPR** — health data is special-category data; you need a lawful basis, ethics approval, informed consent where required, de-identification, data-minimisation, and a data protection impact assessment.
- Research ethics committee approval for any patient data or tissue (extracted teeth count as human tissue in many institutions).
- Data-sharing agreements that allow publication and, ideally, public release of a de-identified dataset.

Interoperability

- **DICOM** — the imaging standard; storing OCT and results in DICOM makes integration with practice imaging systems easier.
- **HL7 FHIR** — the standard for exchanging clinical data and results with health records.
- Dental coding systems for tooth numbering (e.g. FDI notation, where “36” means lower-left first molar) so outputs attach to the right tooth.

7. What you need to know

Knowledge checklist

- **Clinical domain** — dental anatomy, caries staging (ICDAS), caries management philosophy (minimally invasive dentistry), restorations, periodontal and oral mucosal basics. Learn how dentists actually make each decision today.
- **OCT physics and artifacts** — speckle, attenuation, surface reflection, polarization contrast, and what saliva, motion, and probe angle do to an image.
- **Medical image analysis** — segmentation and detection architectures, 3D volumes, preprocessing, patient-level splitting.
- **Trustworthy ML** — uncertainty, calibration, OOD detection, selective prediction, domain adaptation, shortcut learning.
- **Clinical evaluation** — diagnostic accuracy statistics, decision curve analysis, reader-study design, reporting guidelines.
- **Human factors** — interface design for clinicians, automation bias, alert fatigue.
- **Regulation and governance** — SaMD, MDR, EU AI Act, ISO 14971, IEC 62304, GDPR, ethics approval.

- **Engineering** — reproducible pipelines, versioned data and models, DICOM handling, monitoring.
- **Collaboration** — a clinical partner who owns the decision and ground truth; an imaging partner who owns the device.

Common pitfalls

- Building a model before defining the clinical decision it supports.
- Splitting data by image rather than by patient or tooth (leakage and inflated results).
- Evaluating only on the same device and site as training.
- Reporting accuracy without calibration, uncertainty, or failure analysis.
- Relying on heatmaps as the only explanation.
- Ignoring the quality gate, so the system confidently analyses unusable scans.
- Treating ex vivo results as proof of in vivo performance.
- Overclaiming autonomy; the defensible framing is clinician support.

Glossary

Term	Meaning
CDSS	Clinical decision support system
Intended use	Formal statement of who uses the system, for what decision, on which patients
Hybrid CDSS	ML perception plus explicit clinical rules
Quality gate	Step that rejects unusable scans before analysis
Abstention / selective prediction	System declines to answer when unreliable
OOD	Out-of-distribution: input unlike the training data
Calibration	Agreement between stated confidence and actual correctness
Decision curve analysis	Method measuring clinical net benefit of acting on predictions
Reader study	Experiment comparing clinicians with and without the tool

Term	Meaning
Automation bias	Tendency to follow system output even when wrong
Alert fatigue	Clinicians ignoring alerts that fire too often
SaMD	Software as a Medical Device
MDR	EU Medical Device Regulation
Notified body	EU organisation that certifies medical devices
ISO 14971 / IEC 62304	Risk management / medical software lifecycle standards
DICOM / HL7 FHIR	Imaging standard / clinical data exchange standard
FDI notation	Two-digit tooth numbering system
Data drift	Change in incoming data after deployment that degrades performance